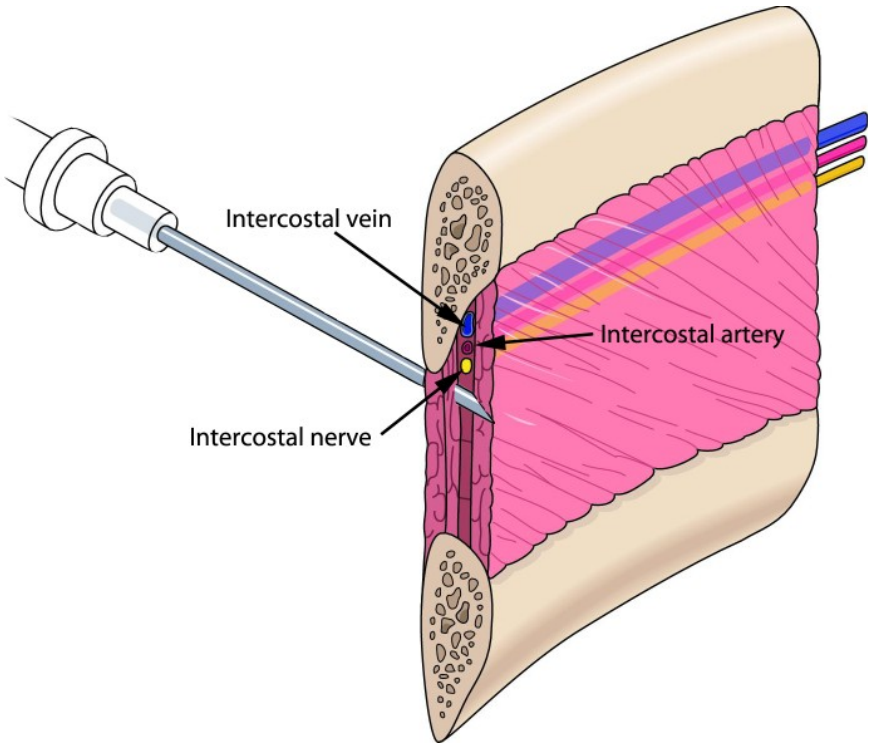
Intercostal nerve blocks are typically performed for postoperative pain management for thoracic or upper abdominal surgery or to relieve pain associated with cancer, fractured ribs, or shingles. The intercostal nerves originate from the dorsal and ventral rami of the thoracic spinal nerves and travel from the intervetebral foramen along the underside of the rib.

As the picture below indicates, the intercostal neurovascular bundle consists of the vein, artery, and nerve with the vein being most superior and the nerve being most inferior.



The block is performed by placing the patient in either the supine or lateral decubitus position. The correct rib is identified by palpation and a skin wheal is raised with local anesthetic at the inferior border of the rib. A 22 to 25 gauge needle is walked off the inferior forder of the rib and advanced about 0.5 centimeters under the rib. After negative aspiration for blood and air, 3-5 mL of local anesthetic is injected.

It should be noted that intercostal blocks generated the highest blood levels of anesthetic per volume injected and the anesthesia provider must be careful to avoid administering toxic levels. Any indication that the pleura has been compromised should be evaluated with a chest xray.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1018-1019.

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